

KISHORE KUMAR A

☎ [+91 9600025176](tel:+919600025176) | ✉ kishorekumar020413@gmail.com | 💼 [LinkedIn](#) | 💻 [GitHub](#) | 🌐 [Portfolio](#)

PROFESSIONAL SUMMARY

Final-year EIE student at Anna University with proven experience in Embedded Hardware Design, Industrial Automation, and IoT-based sensor interfacing system development through industrial training and multidisciplinary academic projects. Hands-on expertise in STM32, ESP32, Siemens S7-1200 PLC, Factory I/O, and industrial communication protocols.

EDUCATION

Madras Institute Of Technology Chrompet,
Anna University – Chennai, Tamil Nadu

2023- Present

Bachelor of Engineering in Electronics and Instrumentation Engineering

CGPA: 7.39 / 10 (Upto 6th semester)

Relevant Coursework & Knowledge: Industrial Instrumentation, Sensors And Transducers, Industrial Automation, Embedded System, Process Control, Analog & Digital Electronics.

TECHNICAL SKILLS

- **Programming** C, Python, PLC Programming (Basics)
- **Hardware & Platforms:** Siemens S7-1200 PLC, 8051, STM32, ESP32, Blynk IoT
- **Communication Protocols:** HART, Modbus, PROFIBUS, FOUNDATIONAL Fieldbus, CAN, UART, I²C, SPI
- **Software Utilities:** MATLAB, TIA Portal, Factory I/O, LabVIEW, LTspice, Proteus, Fusion 360

INTERNSHIP EXPERIENCE

L&T Technology Services, Chennai | Embedded Hardware Design Intern | May 2026 – Jun 2026 | [Link](#)

- Completed 5 weeks of industrial training in Embedded Hardware Design, gaining exposure to the complete **hardware development lifecycle**, including **schematic design, PCB assembly (THT & SMT), hardware validation, and testing methodologies**.
- Applied engineering concepts through **Design Analysis Report (DAR)**, component selection, PCB defect analysis, **PC-based sensor interfacing, DC-DC converter design, and generation of BOM and Netlists**, while understanding industry-standard embedded hardware development practices.

PROJECTS

Digital Twin for Level Process | DEC 2025 – APR 2026 | [Link](#)

- Developed a **PLC-based digital twin** of an industrial level control process using **Siemens S7-1200, TIA Portal, and Factory I/O**, implementing **PID control, analog signal scaling, and real-time synchronization** between the PLC and virtual process.
- Designed an **HMI for real-time process monitoring and control, integrating response trend plotting** to visualize level variations and validate closed-loop process performance in an Industry 4.0-based automation environment.

IoT-Based Soil Moisture Monitoring & Pump Control | OCT 2025 – NOV 2025 | [Link](#)

- Designed an **ESP32-based smart irrigation system by integrating a soil moisture sensor, relay-controlled water pump, and Blynk IoT** to provide real-time soil moisture visualization through a mobile dashboard.
- Implemented threshold-based automatic irrigation control that activates the pump below the predefined soil moisture level and stops it upon reaching the target moisture, enabling remote monitoring and efficient water management.

ONGOING PROJECT

Smart Crop Prediction and Monitoring System using STM32 & ESP32 | Present |

CMRG Government Funded project:

- Developing a **smart agriculture platform using STM32, ESP32, and an 8-in-1 soil sensor to acquire real-time soil and environmental parameters**, with long-range wireless data transmission and cloud-based remote monitoring.
- Integrating a **Machine Learning model to predict suitable crops and generate fertilizer recommendations based on soil characteristics**, while enhancing the existing prototype into a scalable precision agriculture solution through intelligent decision support.

MINI PROJECTS

Automated Assembly and Quality Check Station | FEB 2026 – APR 2026 | [Link](#)

- Simulated an **industrial assembly and inspection process using Factory I/O and Control I/O**, implementing automated assembly, sensor-based quality verification, reject handling, safety interlocks, and production counters for efficient process automation.

Low Voltage DC Voltmeter | JAN 2025 – APR 2025 | [Link](#)

- Designed a 741 Op-Amp-based Voltage-to-Current (V-I) converter **to measure low DC voltages across multiple ranges (1 V, 5 V, 10 V, and 13 V)**, and calibrated the circuit using an analog voltmeter to validate accurate voltage measurement.

Maze Solver Robot using Digital Logic | JAN 2025 – APR 2025 |

- Designed and implemented a **maze-solving robot using IR sensors, digital logic gates, and motor driver circuitry to detect obstacles and autonomously navigate** through a predefined path without microcontroller-based control.

CERTIFICATIONS

- **A Brief Introduction to Micro-Sensors** | NPTEL | IISER Bhopal | **Elite Silver Certificate (85%)** and ranked among the **Top 5%** of course participants | [Link](#)
- 3 Days Hands-On Skill Development Programme On **Industrial Automation Using Siemens PLC And HMI** | [Link](#)
- 3 Days Hands-On Skill Development Programme On **Application Of AI In Smart BMS For EV** | [Link](#)

ORGANISATION

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| • ISA- INTERNATIONAL SOCIETY OF AUTOMATION
Student Member. | 2025- Present |
| • YRC- YOUTH RED CROSS, MIT
Volunteer | 2023- Present |
| • IEA- INSTRUMENTATION ENGINEERS ASSOCIATION, MIT
Student Member, Volunteer | 2023-Present |